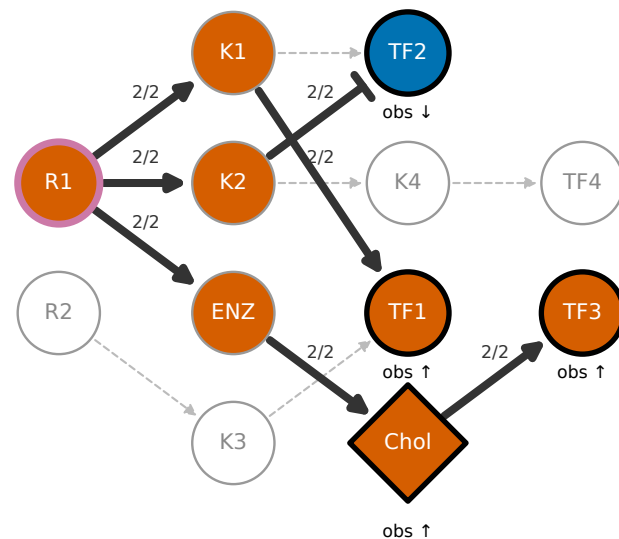
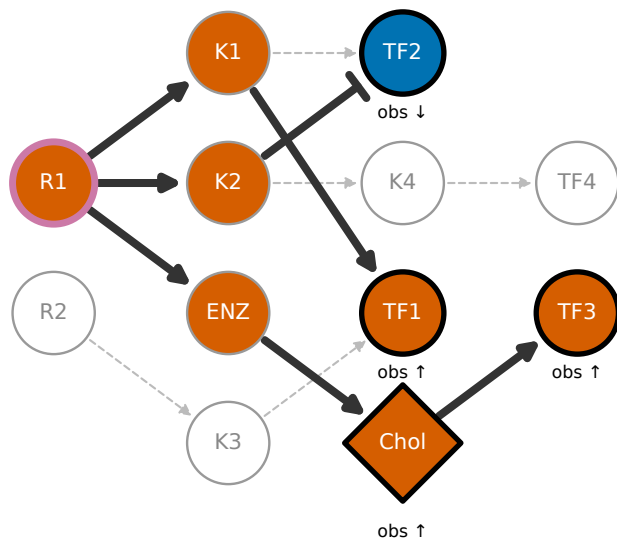
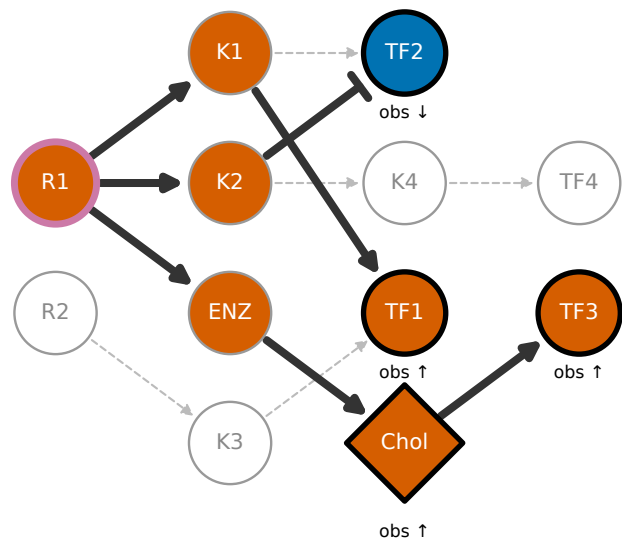


A One sample, input sign given (R1 = +1)

B One sample, input sign inferred (R1 = 0)

C Two identical samples, solved jointly



state +1 (up)
 state 0 / not used
 measured node (obs = observed sign)
 metabolite
 state -1 (down)
 input node
 ——— PKN edge, not selected
 ——— selected edge (→ activation, — inhibition)

Toy PKN: 12 nodes, 12 signed edges (5 decoys); Chol = Metab_HMDB0000067_c (cholesterol, COSMOS-style name). CORNETO 1.0.0rc8 CarnivalFlow, $\lambda = 0.1$, solver highs (limit 60 s); one optimal solution per panel, no sampling. Edge width and n/N in C = fraction of samples selecting the edge. Measured nodes with sign mismatches across panels: 0.